

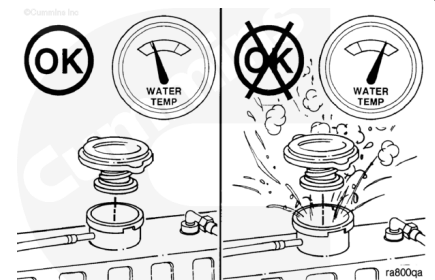
Trainings ▾

Test

Overflow Method

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

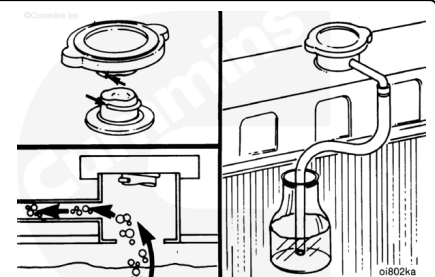


Remove the cooling system pressure cap.

Install a radiator pressure cap that has had the spring and pressure relief valve removed.

The pressure cap **must** make a tight seal.

Attach a rubber hose to the radiator overflow connection. Put the free end of the hose in a container of water.

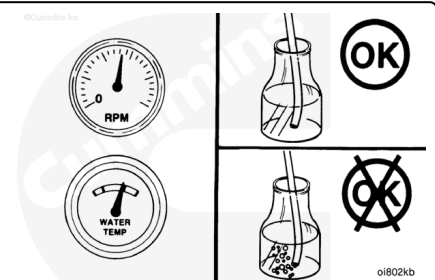


Operate the engine at rated rpm until it reaches a temperature of 85°C [185°F] with the thermostat open.

Check for a continuous flow of air bubbles from the hose in the water container.

A continuous flow of bubbles can be the result of:

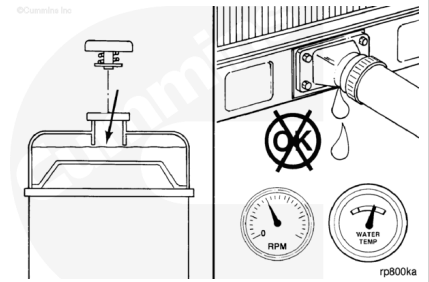
- Air compressor cylinder head leak
- Cylinder head combustion gas leak
- Incorrect cylinder protrusion
- Cracked or broken cylinder liners.



If air is **not** in the system, perform the following:

- Remove the test equipment.
- Check the coolant level and fill if necessary.
- Install the correct radiator pressure cap.

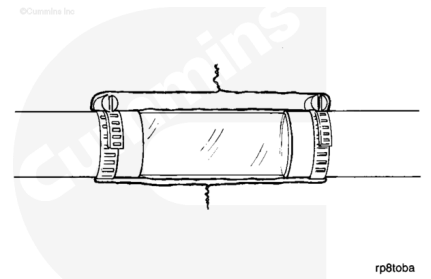
- Operate the engine to a temperature of 80°C [180°F].
- Check for coolant leaks.



Sight Glass Method

Install a clear Pyrex® tube in the water connection between the engine radiator and the thermostat housing.

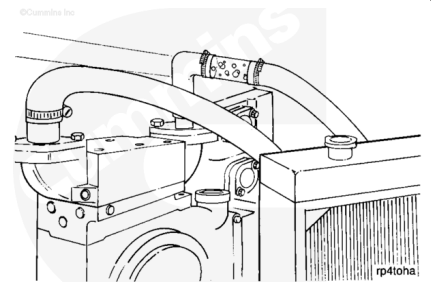
If desired, the Pyrex® tube can be installed in the vent line between the thermostat housing and the radiator.



⚠ WARNING ⚠

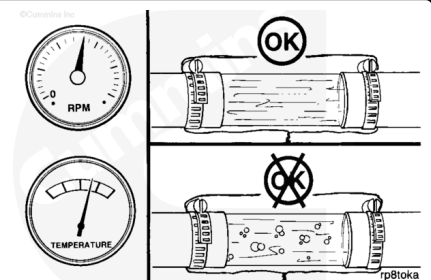
Use wire lace to cover the tube and hose. Personal injury can result if the Pyrex® tube separates from the hoses.

Cover the tube and hose with a wire lace.



Operate the engine at 75 percent of rated rpm until it reaches an operating temperature of 85°C [185°F].

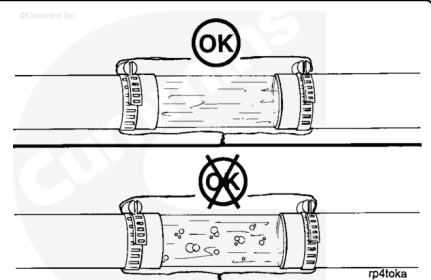
Check for a continuous flow of air bubbles in the hose.



A continuous flow of air bubbles indicates excessive air in the system. Continue to operate the engine for 25 minutes.

If the flow of bubbles does **not** stop, check for the following:

- Air compressor cylinder head leak
- Cylinder head combustion gas leak
- Incorrect cylinder protrusion
- Cracked or broken cylinder liners.



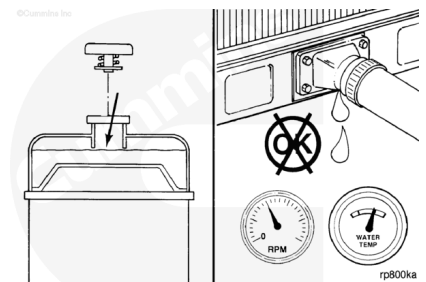
If no air is found in the system or if the flow of bubbles stops within 25 minutes, perform the following:

Remove the test equipment.

Check the coolant level and fill if necessary.

Install the correct radiator pressure cap.

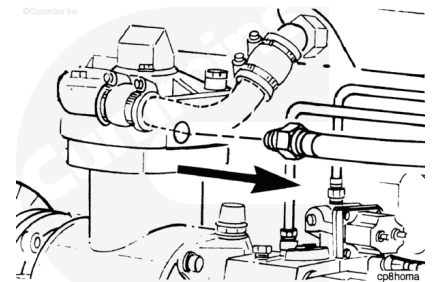
Operate the engine until it reaches a temperature of 85°C [185°F], and check for coolant leaks.



Air Compressor

⚠ CAUTION ⚠

The air compressor discharge must be disconnected at the compressor. This will allow the compressor to discharge air to the atmosphere to prevent the compressor from overheating during this test.



⚠ CAUTION ⚠

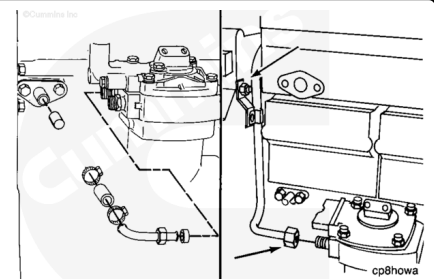
Do not operate the engine over 5 minutes with the components isolated from the cooling system. Component damage can result.

Disconnect the air compressor discharge line.

Disconnect and plug the coolant supply and coolant drain lines, from the air compressor.

Repeat the sight glass method or overflow method test.

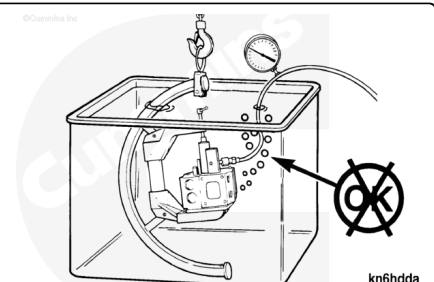
If air is **not** found in the cooling system with the air compressor isolated, replace the air compressor cylinder head gasket. Refer to the Holset Master Repair Manual, Bulletin 3666121.



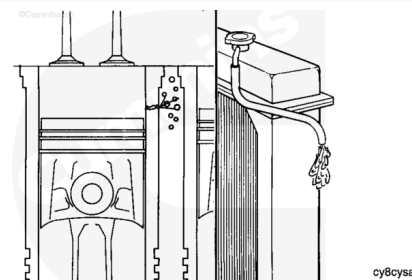
Cylinder Head and Cylinder Liner

A cracked or defective cylinder head will pressurize the cooling system and force the coolant out the overflow tube.

Check the cylinder head gasket. Refer to Procedure 002-004 ([/qs3/pubsys2/xml/en/procedures/56/56-002-004-tr.html](https://qs3/pubsys2/xml/en/procedures/56/56-002-004-tr.html)).

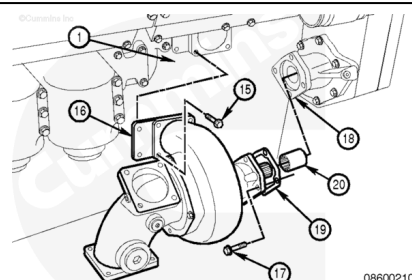


A cracked or porous cylinder liner will pressurize the cooling system and force coolant out the overflow tube.



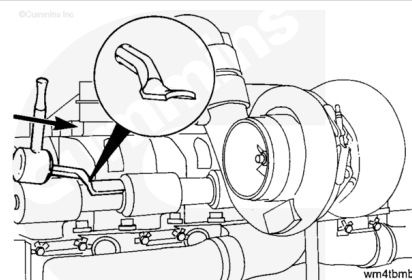
To determine the cylinder liner that is leaking, perform the following:

- Remove the water pump. Refer to Procedure 008-062 (/qs3/pubsys2/xml/en/procedures/56/56-008-062-tr.html).
- Remove the collar.
- Install the water pump without the collar (20).



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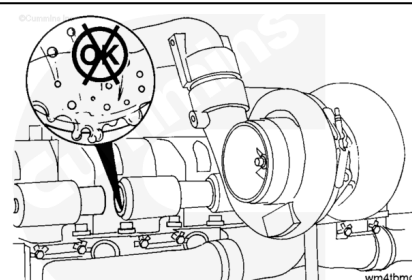
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Drain the coolant down to where coolant is level with the water manifold tubes.

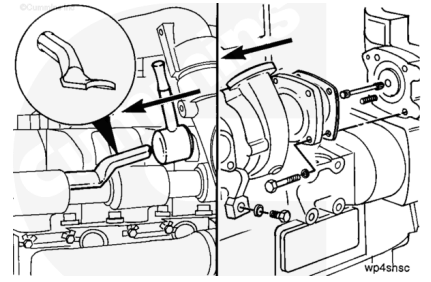
Slide the water manifold tubes into the rocker lever housing.

Operate the engine at low idle for a maximum of 4 minutes. Check inside the water passage of each rocker lever housing for air bubbles. Air bubbles in the coolant indicate a porous or cracked cylinder liner.

A defective cylinder liner **must** be replaced. Refer to Procedure 001-028 (/qs3/pubsys2/xml/en/procedures/56/56-001-028-tr.html).



If bubbles are **not** detected, install the water tubes and water pump collar.



Fill the cooling system. Refer to Procedure 008-018
(</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>).

